BABITA AWASTHI

Full-Stack Software Developer

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PROFESSIONAL SUMMARY

Full-Stack Software Developer with over 2 years of experience in API development, system integration, unit testing and debugging. Builds and deploys production web applications using TypeScript, Next.js, NestJS, Python (FastAPI) and PostgreSQL, with Git-based workflows and CI/CD. Experienced in REST API design, role-based access control, multi-tenant systems, automated testing, and QA and code review for municipal government software.

PROFESSIONAL EXPERIENCE

Full Stack Developer (Full-time) | Data Construct, Lalitpur **Feb 2024 – Present**

- Primary contributor to live exam platform prepbox.net, an **88-endpoint REST API** serving **50+ institutions**.
- Designed a dynamic-programming algorithm that generates a unique question paper per attempt with exact total marks: **5,000 questions in under 500 ms**, covered by **36 unit tests**.
- Built exam engine APIs with per-student shuffled papers, autosave and auto-submit, preventing answer sharing.
- Built analytics dashboards with PostgreSQL aggregation, and Excel/CSV bulk import with row-level validation.
- Built **20+ reusable dashboard components** in Next.js and TypeScript, and delivered frontends for an enterprise sports management platform.
- Performed QA testing and code reviews for project management systems used by **2 municipal governments**.

Freelance Full-Stack Developer (Part-time) | Client: Universal University Group **Apr 2026 – Present**

incountryrepresentative.com | globalsathi.com | interviewforyou.com

- As **sole developer**, built a multi-tenant study-abroad SaaS: an agency CRM (one subdomain per agency) and a central admin platform on one **NestJS/PostgreSQL REST API** (Prisma, ~30 data models), replacing paper, spreadsheet and email workflows.
- Designed role-based access control (4 roles, 9 permissions) from one shared map used by API and UI, verified by **276 automated tests** (121 end-to-end, 121 route-coverage, 34 frontend) that fail any unprotected route.
- Built an **eligibility engine** filtering the course catalog by GPA, study gap, qualification and English-test bands, with course-level overrides.
- Built application tracking: a 4-step draft wizard, versioned documents, and **29 statuses** feeding an **8-stage** pipeline dashboard.
- Replaced per-agency university copies with one **global catalog** and duplicate detection (unique index plus similarity check); a hand-written migration merged duplicates without losing any agency's links.
- Hardened authentication: email OTP, rotating single-use refresh tokens and login lockouts. Moved auth state from Redis to PostgreSQL.
- As **lead developer**, built an AI mock-interview service (FastAPI, pgvector RAG question selection, LLM-as-judge scoring, liveness check). Integrated it with the CRM through server-to-server APIs, Deepgram, Cloudflare R2 and AWS SES, with **460+ offline pytest tests**.
- Deployed with Docker on Fly.io through **GitHub Actions** (test, build, deploy, smoke check) with per-pull-request preview environments.

PERSONAL & CLIENT PROJECTS

- **Graha Mantra** (grahamantra.vercel.app): multilingual Jyotish web platform for a client, built with React and Supabase, deployed on Vercel and in active use.
- **Online Flat Rental Platform** (GitHub): real-time chat with Socket.io, OTP and JWT authentication.
- **AthletiCore** (GitHub): live powerlifting competition app with real-time leaderboards and QR check-in.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

API Development & Integration: REST APIs, NestJS, Express, FastAPI, JWT, RBAC, third-party service integration

Frontend: React, Next.js, Tailwind CSS, React Query, Redux Toolkit

Databases: PostgreSQL, Prisma, TypeORM, SQLAlchemy, pgvector, MongoDB

Testing & Debugging: Unit and integration testing, Jest, Vitest, pytest, code review

Version Control & DevOps: Git, GitHub Actions, Docker, Linux, Fly.io, AWS SES, Cloudflare R2

EDUCATION

B.E. Computer Engineering | NCIT, Pokhara University, Lalitpur | CGPA: 3.41 / 4.0 Aug 2025

CERTIFICATIONS

AWS Academy Graduate: AWS Cloud Foundations 2024

LANGUAGES

Nepali (Native) | English (Fluent)